

TIC 96818680

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 3.239468 d (ALIASSED, see below)
-> corrected period: 6.478936 d
Epoch (t0) 3098.4329 BTJD
Duration 0.0580 d
Depth 0.439 % (4,392 ppm)
SNR 113.9

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

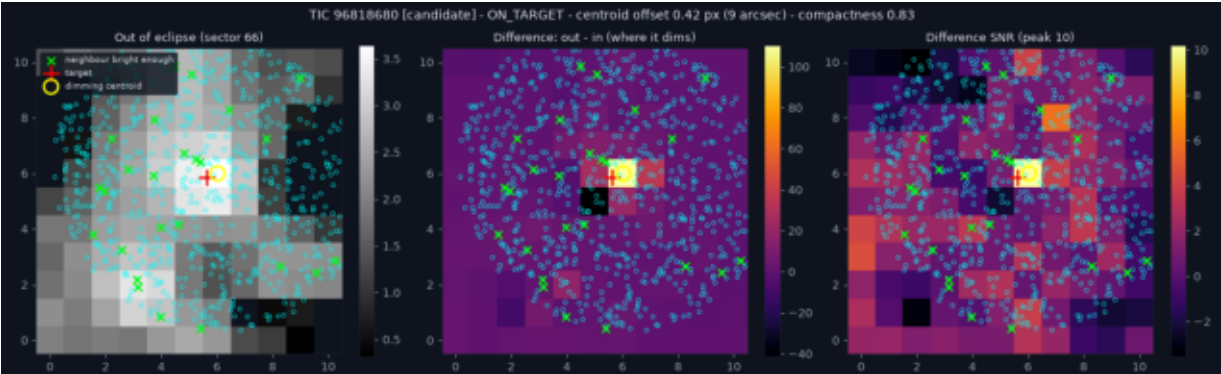
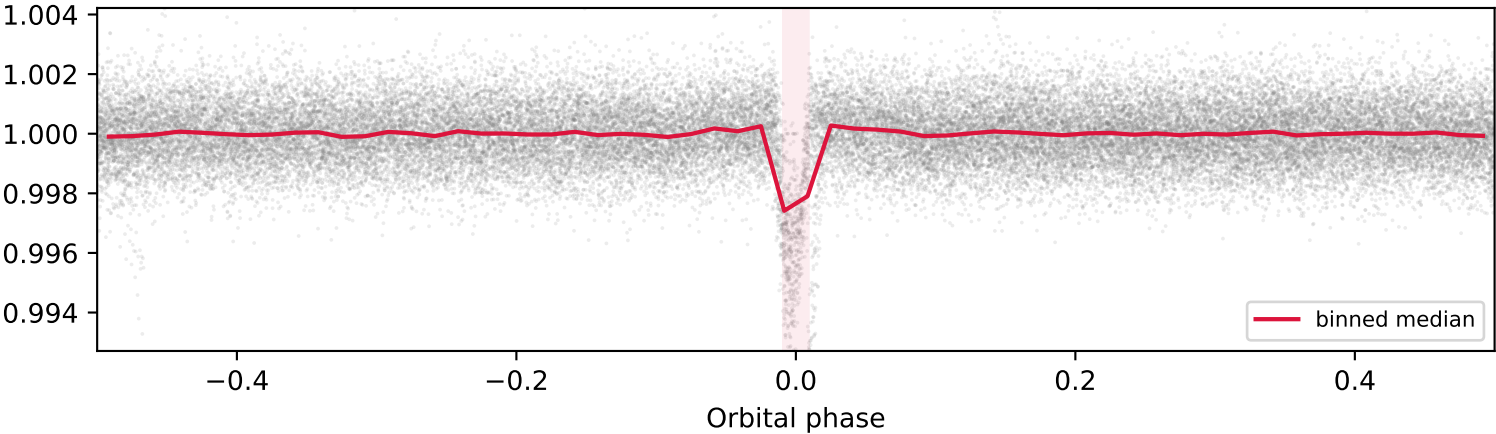
Eclipses observed 17 (gate >= 3)
Robust depth ratio 0.84 (gate >= 0.50)
Robust depth sigma 75.5 (gate >= 7.0)
Odd/even z-score 4.77 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 66 light-curve dip in this sector: 87.1 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 10.19 (gate >= 5.0) Compactness 0.83 (gate >= 0.50)
Dimming centroid offset 0.42 px (9 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 96818664
0.75 px away, dTmag +5.66 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.77 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 6.478936 d, in any write-up.